



Guide to Documentation for DesignWare SDRAM and Static Memory Controller



DesignWare IP Component Document Set for DW_memctl

The DesignWare SDRAM and Static Memory Controller (DW_memctl) memory controller that you can configure and synthesize in order to control synchronous DRAMs – SDR-SDRAM, Mobile SDR-SDRAM, Mobile DDR-SDRAM, DDR-SDRAM – as well as Static memories – SRAMs and FLASHes.

The DW_memctl is part of a suite of components that handle memory management, interrupt control, intra-bus and external I/O, timer functions, reset/watch dog functions, and more. The coreAssembler tool enables you to easily connect, test and synthesize your bus system, no matter how complex.

Installation information for the DW_memctl is available in the *DesignWare SDRAM and Static Memory Controller Installation Guide*. For release note information, including global issues, refer to the *DesignWare SDRAM and Static Memory Controller Release Notes*.

Table 1-1 lists the documents related to using the DesignWare SDRAM and Static Memory Controller environment, components, and tools.



Note

The links below access only top-level documents and are for installed components. If you would like to view documentation that you cannot access from this table, refer to the following web site:

<http://www.designware.com>

Table 1-1 DW_memctl Document Descriptions

Document Name	Filename	Description
DesignWare SDRAM and Static Memory Controller Databook	dmctl_db.pdf	Describes the DW_memctl (Memory Controller) synthesizable IP, which is used to control synchronous DRAMs (SDRAMs) and Static memories.
DesignWare SDRAM and Static Memory Controller Installation Guide	dmctl_install.pdf	Explains how to download and install the DesignWare components for AMBA and related tools. Lists component and tool versions. Lists environment variables and licenses.
DesignWare SDRAM and Static Memory Controller Release Notes	dmctl_rn.pdf	Contains release note information for all DesignWare SDRAM and Static Memory Controller .
DesignWare DW_memctl Driver Kit User Guide	dw_memctl_driver.pdf	Allows programming DW_memctl Includes APIs and example code. Written in portable C (ANSI C99) for use with any 32-bit processor.

Table 1-2 lists additional documentation related to using the DW_memctl.

Table 1-2 Related Documentation

Document Name	Filename	Description
Verification IP Documents – master, slave, and monitor IP for testbench verification and compliance		
DesignWare AHB Verification IP Databook	ahbvipdb.pdf	Provides information for installing, configuring, and using the AHB master, slave and monitor verification IP.
Release Notes	amba_rn.pdf	
DesignWare APB Verification IP Databook	apbvipdb.pdf	Provides information for installing, configuring, and using the APB master, slave and monitor verification IP.
Release Notes	amba_rn.pdf	
DesignWare Verification IP Tutorial	dwvptut.pdf	Helps you learn about AMBA verification IP models, testbenches, and design simulation.
DesignWare AXI Verification IP User Manual	axi_um.pdf	Provides introductory, usage, and reference material about the AMBA 3 AXI interface models.
VMT Documentation – base technology required for all AMBA Verification IP components		
VMT Installation Guide	vmt_install.pdf	Contains version information, and information on installing and licensing VMT-based verification models, tools.
VMT User's Manual	vmtum.pdf	Contains information on common VMT-based verification models, common command reference, and common configuration information.
Release Notes	vmt_rn.pdf	

Table 1-2 Related Documentation

Document Name	Filename	Description
Application (Tool) Documents		
Using DesignWare Library IP in coreAssembler	coreassembler_tutorial.pdf	Describes the features provided by using DesignWare Library components in coreAssembler. Contains a number of tutorials that take you through the steps of assembling single-layer and multi-layer DesignWare subsystems for AMBA.
coreConsultant User Guide	coreconsultant_user.pdf	Provides information about how to perform configuration, simulation, and verification tasks with reusable synthesizable IP. All individual Synopsys synthesizable IP use coreConsultant.
coreAssembler User Guide	coreassembler_user.pdf	Provides information about how to perform connection, configuration, simulation, and verification tasks with multiple synthesizable IP. Helps to integrate synthesizable IP into simple and complex bus sub-systems.
coreTools Release Notes	coretools_relnotes.pdf	Version and notes for coreTools documentation set (includes coreConsultant and coreAssembler).
Synopsys Common Licensing (SCL) Documentation		
http://www.synopsys.com/keys	HTML pages	Helps you set up a licensing server and license key files to use Synopsys tools and DesignWare libraries. DesignWare license requirements are described in the DesignWare SDRAM and Static Memory Controller Installation Guide .

You can find documentation related to the suite of AMBA 2/AMBA 3 AXI Synthesizable IP components at:

<http://www.synopsys.com/dw/doc.php/doc/amba/latest/intro.pdf>

Web Resources

- DesignWare IP product information: <http://www.designware.com>
- Your custom DesignWare IP page: <http://www.mydesignware.com>
- Documentation through SolvNet: <http://solvnet.synopsys.com> (Synopsys password required)
- Synopsys Common Licensing (SCL): <http://www.synopsys.com/keys>

Getting Help

To obtain support for your product:

- First, prepare the following debug information, if applicable:
 - For environment setup problems or failures with configuration, simulation, or synthesis that occur within coreConsultant or coreAssembler, use the following menu entry:

File > Build Debug Tar-file

Check all the boxes in the dialog box that apply to your issue. This menu entry gathers all the Synopsys product data needed to begin debugging an issue and writes it to the file `<core tool startup directory>/debug.tar.gz`.
 - For simulation issues outside of coreConsultant or coreAssembler:
 - Create a waveforms file (such as VPD or VCD)
 - Identify the hierarchy path to the DesignWare instance
 - Identify the timestamp of any signals or locations in the waveforms that are not understood
- Then, contact Support Center, with a description of your question and supplying the above information, using one of the following methods:
 - *For fastest response*, use the SolvNet website. If you fill in your information as explained below, your issue is automatically routed to a support engineer who is experienced with your product. The **Sub Product** entry is critical for correct routing.

Go to <http://solvnet.synopsys.com/EnterACall> and click on the link to enter a call. Provide the requested information, including:

- **Product:** DesignWare Library IP
- **Sub Product:** Memory
- **Tool Version:** *<product version number>*
- **Problem Type:**
- **Priority:**
- **Title:** DW_memctl
- **Description:** For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood

After creating the case, attach any debug files you created in the previous step.

- Or, send an e-mail message to support_center@synopsys.com (your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
 - Include the Product name, Sub Product name, and Tool Version number in your e-mail (as identified above) so it can be routed correctly.
 - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood
 - Attach any debug files you created in the previous step.
- Or, telephone your local support center:
 - North America:
Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.

- All other countries:
<http://www.synopsys.com/Support/GlobalSupportCenters>

